MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG2DSCECT100 Essential Concepts in Digital Electronics

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt atleast one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	What is the decimal equivalent of the binary number 1011? a) 9 b) 10 c) 11 d) 12	U	CO1
2	What is the output of an AND gate if both inputs are 1? a) 0 b) 1 c) 2 d) Undefined	K	CO2
3	Which of the following gates is known as a universal gate? a) AND b)OR c) NOT d)NAND	K	CO2
4	What is the main characteristic of a flip-flop? a) It stores binary data b) It performs arithmetic operations c) It converts binary data to decimal d) It amplifies signals	K	CO3
5	Which of the following is the Boolean expression for the Sum output of a Full Adder? a) A ⊕ B ⊕ Cin b) (A + B) ⊕Cin c) (A * B) ⊕ Cind) A + B + Cin	K, U	CO3
6	Which of the following is a characteristic of a shift register? a)Store data in sequence b) Perform logical operations c) Convert analog to digital signals d) Amplify weak signals	K	CO3
7	The Karnaugh map (K-map) is used for: a)Simplifying Boolean expressions b) Converting binary numbers c) Generating truth tables d) Analyzing circuit diagrams	U	CO2
8	What is the binary form of the hexadecimal number A? a) 1011 b) 1100 c) 1010 d) 1001	K	CO1
9	Which of the following is the result of the Boolean expression A + $\bar{A}B$? a) Ab) A + B c) A' d) A'B	U	CO2
10	Which Boolean expression is in canonical SOP form? a) A'B + AB' b) A + B c) AB d) A + A'B	U	CO2

1 * 10 = 10

Part B

Short Answer Questions

Answer any**4** Questions

Each question carries 5 marks

11	Explain the operation of a **Half Adder** and a **Full Adder**. Draw their logic diagrams and truth tables.	U	CO3
12	Simplify the following Boolean expression using a Karnaugh Map: $F(A, B, C) = \Sigma(1, 3, 5, 7)$	U	CO2
13	Convert the following numbers: a) Binary 101110 to Decimal b) Decimal 45 to Binary c) Octal 12 to Decimal Explain the steps involved in each conversion.	K	CO1
14	Explain the basic logic gates (AND, OR, NOT) with their symbols and truth tables. Also, show the truth table for a **NAND gate** and explain its operation.	K	CO2
15	Convert the following hexadecimal number to decimal a) E5 b) A85	U	CO1
16	Express the decimal number -39 as an 8 bit number in the sign magnitude. 1's complement and 2's complement	U	CO1

4 * 5 = 20

Part C

Essay Questions

Answer any**2** Questions

Each question carries 10 marks

17	a) Discuss the different types of logic gates, with their truth tables and symbols? b) write the laws and rules of Boolean algebra?	K	CO2
18	Simplify the following Boolean expressions using **K-map** a)A′B′C′D+AB′C′D+ABC′D+ABC′D′ b)A′B′C+AB′C′+AB′C+A′BC′	U	CO2
19	a) Determine the binary number for the following hexadecimal number I) 10A4 II) CF8E III) 9742 b) Convert the following octal number to decimal I) 1024 II) 7765	U	CO1
20	Explain the working principle of aRing Counter**.**	K	CO3

2 * 10 = 20